

Vectors

(Assignment Solutions)

Question 1 :

```
vector<int> findErrorNums(vector<int>& nums) {  
    vector<int> ans;  
    int n = nums.size();  
    vector<bool> isPresent(n+1, false);  
  
    for(int i=0; i<n; i++) {  
        if(isPresent[nums[i]]) {  
            ans.push_back(nums[i]);  
        }  
  
        isPresent[nums[i]] = true;  
    }  
  
    for(int i=1; i<isPresent.size(); i++) {  
        if(!isPresent[i]) {  
            ans.push_back(i);  
            break;  
        }  
    }  
  
    return ans;  
}
```

Note - This solution is brute force & more optimized approaches for the same Qs exist that use Bit Manipulation.

Question 2 :

```
int maxArea(vector<int>& height) {
    int left = 0;
    int right = height.size() - 1;
    int maxArea = 0;

    while (left < right) {
        int currentArea = min(height[left], height[right]) * (right -
left);
        maxArea = max(maxArea, currentArea);

        if (height[left] < height[right]) {
            left++;
        } else {
            right--;
        }
    }

    return maxArea;
}
```

Question 3 :

```
vector<vector<int>> threeSum(vector<int>& nums) {
    vector<vector<int>> ans;
    int n = nums.size();

    sort(nums.begin() , nums.end());

    for(int i=0; i<n; i++){
        if(i>0 && nums[i] == nums[i-1]) {
            continue;
        }

        int j = i+1;
        int k= n-1;
```

```
while(j < k) {  
    int sum = nums[i] + nums[j] + nums[k];  
    if(sum < 0) {  
        j++;  
    }  
    else if(sum > 0) {  
        k--;  
    }  
    else {  
        vector<int> temp = {nums[i] , nums[j] , nums[k]};  
        ans.push_back(temp);  
        j++;  
        k--;  
  
        while(j<k && nums[j] == nums[j-1]) {  
            j++;  
        }  
  
        while(j<k && nums[k] == nums[k+1]) {  
            k--;  
        }  
    }  
}  
return ans;  
}
```